A062 — Gravitational Source and Test-Body Closure

v0.6.0: Time-Resolved Vortex Energy, Conditional Free-Space Green Response and Equivalence Gates

SST-Workbench | source-pinned, hypothesis-separated falsifier

13 October 2026

1 Status, identification and scientific boundary

The original A062-v0.1.0–v0.5.0 results remain separate. Two *independently pinned* E011 trefoil geometries, Gilbert and KnotPlot, initialize a smooth finite Gaussian vorticity tube. A062-v0.5.0 established only short-time *periodic* nonlinear Euler/Navier–Stokes integration, not isolated knot equilibrium. v0.6.0 now tests source integrals and a *conditional* Green-field response for the time-resolved velocity snapshots. The assumed mediator is **not derived** from Euler, nor is universal gravitation. Numerical agreement with $1/R^2$ must **not** promote G8 to PASS.

Automatic run provenance

Field	Recorded value
Run ID	20261013T213545627260Z-b9d63e62
Run phase	BLIND
Protocol bundle SHA-256	e9aa3952bac1811dcfd1affcba30869d4fa81fd092d32435ed9a800f9de5f288
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	2af9e8dd7bd38afebe1f42a47eb2d87254858b6f8d47affa72cf2c340d620ebf

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Framework v1.0.8 protocol frozen and pinned?	Framework v1.0.8 protocol, FREEZE and
G1	PASS	Two original E011 provider SHA-256 verified?	Both E011 source SHA-256 and upstream
G2	PASS	Finite vortex tube energy and solenoidal?	Static finite smooth Gaussian tube has fini
G3	PASS	Nonzero energy-equivalent source, unfitted?	Positive kinetic E/c^2 in specified scale hy
G4	FAIL	SST-73 localized stress gives monopole?	SST-73 free-space stress double divergence
G5	PASS	C++ / Python finite-tube energy parity?	C++17 FP64 agrees with Python for stati
G6	NOT_APPLICABLE	SYCL hardware qualification requested?	Disabled in profile
G7	PASS	Independent E011 static energy agreement?	Two independent E011 source providers re
G8	UNRESOLVED	Physical clock, equivalence and G closed?	No intrinsic Swirl-Clock mediator, physical
G10	PASS	Short-time 3D Euler/NS numerics qualified?	Short-time 3D periodic full Euler/NS scree
G11	UNRESOLVED	Long-term stable slender trefoil proved?	No long-term topology-preserving slender
G12	PASS	Analytic Gaussian Green benchmark passed?	Free-space Green gravity kernel matches in
G13	PASS	Conditional Poisson far-field converged?	CONDITIONAL assumed Poisson Green-fi
G14	FAIL	Local-only clock supports 1/R field?	An instantaneous purely local algebraic Sw
G15	UNRESOLVED	Universal passive test coupling derived?	Passive gravitational charge-to-inertial mas
G16	UNRESOLVED	SI G and clock kinetic action derived?	No physical SI K=G, scalar-action stiffness
G17	PASS	Periodic zero-mode guard passed?	For periodic fields the SST-73 derivative sc
G18	UNRESOLVED	Isolated dynamics and raw data qualified?	No isolated slender vortex time evolution a
G19	PASS	Independent native C++ Green parity?	Independent C++17 kernel reproduces cor
G9	DEFERRED	Reveal authorized and independently done?	Reveal is a separate, commitment-verified

Backend manifest


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      "max_windows_v050_energy_diff": true
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  "The E011 vortex cores are heavily smoothed, periodic, short-lived solver controls",
  "Free-space lift of periodic-cell energy is an assumption, not an isolated Euler/NS solution",
  "G4 FAIL applies to isolated SST-73 double divergence; G14 FAIL applies only to purely algebraic loca
  "G8/G11/G15/G16/G18 unresolved; no first-principles gravitational closure"
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Source manifest


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2 Hydrodynamic active-source candidate

The E011 tube is smoothed with $a = 0.11$ in a dimensionless periodic box $L = 1.2$, whose upstream trefoil geometrical reach is smaller ($\simeq 0.025$–0.028). Evolve the divergence-free velocity via the projected unforced Euler/NS equations,

$$\partial_t \mathbf{u} = \mathbb{P}(\mathbf{u} \times \boldsymbol{\omega}) + \nu \Delta \mathbf{u}, \qquad \boldsymbol{\omega} = \nabla \times \mathbf{u}, \qquad \nabla \cdot \mathbf{u} = 0. \tag{1}$$

The cell energy provides a candidate energy-equivalent mass *only after specifying a physical scale*,

$$\rho_E = \tfrac{1}{2}\rho_f |\mathbf{u}|^2, \qquad Q_E = \int \rho_E / c^2 \, d^3x > 0. \tag{2}$$

For the present numerical source-screen only, set $\rho_f = c = 1$ and use $Q = \int_{\text{cell}} |\mathbf{u}|^2/2 \, d^3x$. This is a normalized *kinetic charge*, not SI gravitational mass. Both providers, both $\nu \in \{0, 0.005\}$, and snapshots $t \in \{0, 0.03\}$ are evaluated; the actual velocity snapshots are regenerated and compared against user-provided Windows v0.5.0 energies.

3 Explicitly postulated gravity mediator

An additional scalar field law is assumed, rather than derived:

$$\nabla^2 \Phi = 4\pi K q, \qquad q = \tfrac{1}{2}|\mathbf{u}|^2, \qquad K = 1 \quad \text{(dimensionless test)}. \tag{3}$$

Then in an *isolated free-space lifted-cell proxy*,

$$\Phi(\mathbf{R}) = -K \int_{\text{cell}} \frac{q(\mathbf{x})}{|\mathbf{R} - \mathbf{x}|} \, d^3x, \tag{4}$$

$$\mathbf{g}(\mathbf{R}) = -K \int_{\text{cell}} \frac{q(\mathbf{x})(\mathbf{R} - \mathbf{x})}{|\mathbf{R} - \mathbf{x}|^3} \, d^3x. \tag{5}$$

By construction, when $Q \neq 0$, $\Phi \sim -KQ/R$ and $|\mathbf{g}| \sim KQ/R^2$. **This slope is imposed by the field equation.** The periodized underlying Euler/NS field has not become an isolated solution by copying one cell into this integral. An independent C++17 numerical kernel and the closed-form Gaussian reference

$$|g(r)| = \frac{KQ}{r^2} \left[\operatorname{erf}\left(\frac{r}{\sqrt{2}\sigma} \right) - \sqrt{\frac{2}{\pi}} \frac{r}{\sigma} e^{-r^2/(2\sigma^2)} \right] \tag{6}$$

certify the conditional numerical Green calculation only.

4 Competing and negative source controls

SST-73 postulates $S_C = \lambda \partial_i \partial_j (\rho_f u_i u_j)$. For localized smooth stress with adequate decay, $\int_{\mathbb{R}^3} S_C \, dV = 0$; on a periodic domain its zero Fourier mode also vanishes. The SST-73 *local stress monopole* hypothesis therefore remains G4 FAIL, despite vortex-tube energy being positive.

A *purely local instantaneous* clock map $S_{\text{loc}}(\mathbf{x}) = \sqrt{1 - |\mathbf{u}|^2/c_{\text{ref}}^2}$ returns to a constant outside compact support and cannot by itself make a long-range monopole. G14 FAIL applies **only** to that narrowly specified local map, not to an independent propagating Swirl-Clock mediator. A periodic Poisson problem with a strictly positive mean source requires background subtraction; its neutralized solution is not identical to the isolated free-space calculation.

5 Gravitational test-body and coupling gates

The source field does not by itself establish the acceleration of a different material knot. With passive charge q_B and inertial mass m_B,

$$\mathbf{F}_B = q_B \mathbf{g}_A, \qquad \mathbf{a}_B = \frac{q_B}{m_B}\mathbf{g}_A. \tag{7}$$

The universality $q_B/m_B = 1$, the SI scale $K = G$, a kinetic/action term for the clock mediator, the fully isolated dynamical limit, and a gram-scale microgravity comparison are **not obtained** from these calculations. They remain G15/G16/G18 UNRESOLVED and keep G8 UNRESOLVED. The original Canon v0.8.35 G normalization includes the orthodox Planck time, so using it as an independent derivation is excluded.

6 Predeclared numerical thresholds and gate discipline

G12: Gaussian manufactured relative force error < 0.002. G13: maximum $|n - 2| < 0.03$, maximum relative monopole-amplitude error < 0.025 at $R/L \geq 4$, snapshot energy-integral error $< 10^{-12}$, and v0.5.0 Windows energy-reference mismatch $< 10^{-9}$. G19: independently compiled C++17 Green-gradient parity relative error $< 10^{-10}$; compiler missing gives UNRESOLVED, not PASS. G17 verifies the derivative-source zero mode and periodic-neutrality restriction; G14 is a restricted physical FAIL. No failed gate prevents the remaining independent gates from executing. G10 retains the prior short-time Euler/NS solver control and G11 remains unresolved.

7 Reproducibility and provenance

The original v1.0.8 `LOCAL_VALIDATED_CANDIDATE` framework is included unmodified, with instance-local nonce initialization *only before FREEZE* and unique run-ID exports. Private 32-byte nonce/HMAC files are excluded from released source ZIPs. The known v1.0.8 provisional-manifest hash PDF ordering discrepancy is not repaired by modifying the framework. Both E011 raw-geometry hashes and prior Windows reference snapshots are included and provenance labeled. Public reference papers are not hidden ground truth.

8 References and limitations

SST-18 (Euler near-field vs mediator); SST-34 (direct pressure deprecated as gravity); SST-47–49 (conditional scalar-field EFT); SST-50 (source–test universality requires a dynamical bridge); SST-59/61 (energy and topological stability); SST-73 (organized stress double divergence); SST Canon v0.8.35 (clock, stress, density and calibrated G); A062 v0.1.0–v0.5.0 source-authenticated numerical gates. SST-28 is excluded. The announced Navier–Stokes singularity programme is not used as a substitute for the measured solver controls or as evidence of gravitational closure.